

Response to the Editor and Reviewers

Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP

EDITORS REQUESTS

In particular the following editorial thresholds needs to be met,

- Provide a compelling case that climate outweighs human effects on vegetation assemblage dynamics in the Holocene, including a thorough analysis of the results, clearly indicating the variance explained by models.

Response. We have revised both the analysis presentation and the interpretation of model fit. The revised article now reports total adjusted R^2 separately from the relative allocation of explained variation between human activity and palaeoclimate. It also presents the corrected hierarchical variation-partitioning extraction under signed, zero-truncated, and exclude-negative profiles. In the controlled spatial analysis, palaeoclimate had greater relative importance than human SPD in 891 of 1,098 eligible records (81.1%). Across the 1,156 records with an estimable spatial SPD decomposition, mean total adjusted R^2 was 0.352. The controlled temporal analysis showed the same broad ordering in 69 of 70 estimable region-age comparisons.

Changes in the revised article. We report these results in the main Results section “H1: Holocene human impact secondary to palaeoclimate” and Figs. 3–4. Supplementary Result validation presents the distributions of adjusted R^2 , fit-importance diagnostics, signed results, unique fractions, and the comparison of HVarPart reporting profiles. The Discussion now states explicitly that relative allocations partition the explained component of a fitted model and are not the proportion of total pollen variation explained.

- Justify the usage of data from multiple scales and fully discuss the limitations of your methods, particularly as raised by reviewer #2.

Response. We have bounded the inference to continental and subcontinental patterns in pollen assemblage properties from 8.5 to 2.0 ka BP and now explain the spatial scales represented by each data source. Pollen source areas vary among sites; palaeoclimate is extracted at record coordinates; archaeological radiocarbon dates are collected within a 250

km neighbourhood with a 500 km fallback; and expert-derived human events are regional and record based. No single spatial support can make these sources equivalent. We therefore added explicit discussion of scale mismatch and matched sensitivity analyses for SPD radius and human-proxy definition. We also expanded the treatment of pollen-count and productivity biases, climate-model smoothing and point extraction, archaeological sampling bias, event coarseness and circularity, fixed modern climate zones, and residual spatial and temporal dependence.

Changes in the revised article. The relevant additions appear in Methods under “Data filtering,” “Spatial units for analysis,” “Detection of past human activities,” “Palaeoclimate,” and “Numerical analysis of hypothesis 1”; in the scale- and proxy-limitation paragraphs of the Discussion; and in “Supplementary Methods: Result-validation analyses” and “Supplementary Results: Result validation.”

- Clearly explain your methods in detail.

Response. We substantially expanded the Methods while retaining the structure of the submitted article. The revised text now distinguishes the preparation pool, original H1 model inventory, and smaller analysis-specific cohorts; explains the 25-grain minimum, normal 150-grain rarefaction target, and record-specific lower counts; defines all ten PAPs and their expected correlations; describes the temporal PAP trajectories as equal-weighted averages of record-level posterior predictions over observed coverage; gives the construction and limitations of both human proxies and the palaeoclimate predictors; and distinguishes total model fit, signed hierarchical contributions, zero-truncated display compositions, unique fractions, and shared commonality. The time and spatial controls and all supplementary validation analyses are also described explicitly.

Changes in the revised article. These additions are incorporated throughout the revised Methods, with the detailed quality-control and sensitivity procedures grouped in “Supplementary Methods: Result-validation analyses.” The associated cohorts, model profiles, figures, and tables are identified in their captions and in the Supplementary Results.

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author)

Review of the manuscript

“Climate outweighs human effects on vegetation assemblage dynamics in the Holocene” by Felde et al.

The manuscript submitted by Felde et al. presents a substantial contribution to the field of paleoecology. The study is based on a large dataset of pollen records spanning all continents to investigate the relative influence of climate and human activity on vegetation compositional changes throughout the Holocene. The authors use modelled climate data and archaeological radiocarbon dates, along with expert knowledge on human activities derived from pollen records, to disentangle the effects of climate and human activities on vegetation/pollen composition and diversity. The analytical framework is well developed, employing state-of-the-art statistical methods to explore large-scale pollen composition and diversity changes. The use of Pollen Assemblage Properties (PAPs) to assess compositional changes is innovative and adds a new dimension to paleoecological interpretation. Overall, this is an important and ambitious manuscript that I highly recommend for publication.

My major comments on the manuscript are listed below.

Response. We thank the reviewer for this positive assessment and for the detailed suggestions. We have revised the article to clarify its pollen-domain, temporal, and spatial scope; to report model fit and controlled analyses more fully; and to make the methodological limitations and validation analyses more transparent.

Scope and framing of the study period

The title suggests an analysis of the entire Holocene, yet the dataset ends approximately 2000 years ago. This period excludes the last two millennia, during which anthropogenic impacts on vegetation intensified, including major deforestation events and agricultural expansion. The title should be revised to more accurately reflect the temporal scope of the data and findings. Furthermore, some statements in the discussion, such as the caution against the narrative that human transformation is always destructive, may appear strong given the temporal cut-off at 2 ka BP. I suggest to soften or contextualize this statement to avoid misinterpretation.

Response. We agree. The title now gives the precise 8.5–2.0 ka BP interval and refers to pollen assemblage dynamics rather than vegetation assemblage dynamics. We use the same interval consistently throughout the article. We have also removed the broad claim about human transformation being non-destructive. The revised Discussion states that our result is limited to the represented interval and aggregation scales, does not imply an absence of human transformation, and cannot be extrapolated to the last two millennia or the industrial era.

Changes in the revised article. The new title is “Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP.” The Abstract, hypotheses, Results, Discussion, and Conclusion now use this interval and scale-bounded interpretation consistently.

Use and justification of climate model outputs and human activity proxies

While the use of paleoclimate model data is justified by the limited spatial coverage of proxy-based climate reconstructions, a brief discussion of the benefits and limitations of using model outputs (e.g., smoothed trends and difficulties to simulate climate in some regions) should be included. This will help the readers to better understand the implications using climate model outputs.

Response. We agree and have added this discussion. The revised article explains that the CHELSA-TraCE21k simulations provide spatially complete, physically coherent covariates across the study interval, while their grid resolution, model assumptions, temporal smoothing, and point extraction at pollen-record coordinates can understate local climatic heterogeneity or introduce elevation and source-area mismatches. We consequently interpret climate as the explanatory group represented by these covariates, not as an exact reconstruction of conditions within each pollen source area.

Changes in the revised article. This justification and its limitations are now stated in Methods under “Palaeoclimate” and in the Discussion paragraph beginning “Palaeoclimate simulations provide spatially and temporally complete covariates...”.

In a same way, the use of summed probability densities (SPDs) from archaeological radiocarbon dates is appropriate but should be more critically discussed, although the authors already mention some of the issues. These can underestimate human presence, especially in regions or periods with sparse data. Furthermore, the expert-derived “human-impact events” from pollen data introduce a degree of circular reasoning as it is based on pollen information. The authors should mention the pros and cons of using such approach.

Response. We agree. SPD is now described as an index of changing dated archaeological activity rather than a direct population census. We discuss preservation, calibration-curve structure, archaeological research intensity, sparse regional coverage, and spatial mismatch. We retain SPD as the primary human proxy because it is independent of the pollen response and broadly available. Expert-derived events are retained only as a complementary proxy-sensitivity analysis: they include locally informed evidence not present in the SPD, but are coarse, geographically uneven, incomplete, and partly circular because their definition and the PAP responses can derive from the same pollen record.

Changes in the revised article. These qualifications are now explicit in Methods under “Detection of past human activities,” in the Discussion, and in Supplementary Result validation. The article states: “We therefore interpret the event chronologies as a complementary sensitivity analysis of proxy choice, not as independent validation of the SPD result.”

Background information and previous studies

The introduction would benefit from presenting a bit more prior studies that used similar approaches, e.g., Kuosmanen et al. (2018), Marquer et al. (2017), and Reitalu et al. (2015). These studies provide an important context for methods, data types, and interpretations of vegetation change drivers, although for Europe and regional scales only. Their inclusion would strengthen the rationale for the current study.

Response. We agree that the regional variation-partitioning literature provides useful context and have strengthened that context in the Introduction and Discussion. At the same time, we clarify the difference in response domain: the present study evaluates ten summary properties of pollen assemblages rather than estimating taxon-specific vegetation composition. This distinction is important when comparing our broad-scale inference with regional studies.

Changes in the revised article. The Introduction now places the PAP approach alongside prior regional work, and the Discussion contrasts our multivariate PAP framework with taxon- and land-cover-based reconstructions. We will retain the reviewer-suggested references that directly support this comparison in the final bibliography.

Interpretation of hypotheses and results

Hypothesis 1 states that if human impact is an important driver, its influence should match or exceed that of climate. I would say “the major driver” rather than “important driver” in the case it exceeds climate.

Response. We agree and have recast H1 using “primary driver,” which expresses the intended comparison without implying that a driver must be unimportant whenever it is secondary to climate.

Changes in the revised article. H1 now reads: “The first hypothesis (H1) proposed that human impact was the primary driver of changes in PAPs at continental and subcontinental scales between 8.5 and 2.0 ka BP.”

The authors conclude that climate dominates vegetation change at large spatial scales rather than local scales. This aligns with prior studies. I agree with the authors and clearly support the fact that at finer spatial resolutions, human impacts become more visible and potentially dominant. It is important to mention that the use of PAPs may homogenize vegetation signals and amplify climate-related patterns. This should be acknowledged.

Response. We agree and now distinguish broad-scale coherence from local importance throughout the interpretation. The revised article explicitly acknowledges that aggregating community-level PAPs across continental and subcontinental units can mask intense local

change and favour spatially coherent climatic signals. We also avoid treating PAPs as direct measures of vegetation abundance or taxon-specific response.

Changes in the revised article. The Discussion now states that local human effects may be consequential yet poorly aligned with the spatial footprint of the SPD, pollen-source area, or aggregation units, and that our results should not be used to dismiss well-supported local histories of human transformation.

The manuscript would be greatly improved if the authors provide a discussion of vegetation changes themselves, and how specific taxa or assemblage types respond to climatic or anthropogenic drivers. Such discussion would help ecological interpretations.

Response. We agree that taxon-specific responses are ecologically important, but we have not added a new taxon-level analysis because it would require a different response dataset and inferential design. Our response variables are PAPs calculated from globally heterogeneous pollen records; they do not support attribution of the fitted human or climate contribution to a particular tree, grass, crop, or weed. We have instead made this inferential boundary explicit and expanded the discussion of what PAPs can and cannot represent.

Changes in the revised article. The Discussion now states that the reduced-response sensitivity remains a community-summary assessment and cannot identify taxon-specific responses. It also explains that a taxon-specific reconstruction would require harmonised taxon-level variables, region-specific predictors, and additional validation and is therefore outside the tests performed in this article.

Data representativeness and site characteristics

The choice of 1262 pollen sequences is impressive, but more information is needed about the selection criteria. I would have assumed that much more sites would be available at such spatial scales.

Response. We have expanded the description of record selection and now distinguish the successive cohorts. The preparation and expert-screening pool contains 1,270 records; the original H1 model inventory contains 1,262 records after baseline processing; and each controlled or sensitivity analysis uses a smaller eligible or matched cohort determined by the availability and estimability of its response, predictor, and control variables.

Changes in the revised article. Cohort definitions and filtering criteria are now given in Supplementary Methods and Supplementary Table S1, and every analytical caption identifies the cohort used rather than applying the 1,262-record figure to all analyses.

Site characteristics (e.g., basin size, proximity to human settlements) can significantly influence pollen source area and the sensitivity to land use signals. This aspect, although mentioned by

the authors, should be discussed a bit more in details, especially when interpreting local vs regional patterns.

Response. We agree. We now discuss basin characteristics, variable pollen-source areas, and the mismatch between local land-use signals and the broad archaeological neighbourhoods used for SPD estimation. Because consistent basin-size and settlement-distance metadata are unavailable for all records, these characteristics cannot be included as globally complete covariates, and the inference is explicitly limited to broad spatial patterns.

Changes in the revised article. These limitations are described in Methods and in the Discussion paragraphs on local versus broad-scale inference, SPD radius, pollen productivity, and basin effects. We also added a matched 250 km versus 500 km SPD-radius sensitivity analysis.

The analysis assumes stable spatial climate zones. However, climate zones have shifted over the Holocene, which could affect interpretations of vegetation-climate relationships. A brief discussion of this dynamic would be valuable.

Response. We agree that climate-zone boundaries shifted. The modern Köppen–Geiger groups are used only as fixed analytical strata for summarisation and for the subcontinental temporal analyses; they are not interpreted as stable ecological boundaries through time. We now state that any shifting boundary can mix records whose contemporaneous climates differed from the modern label.

Changes in the revised article. Methods under “Spatial units for analysis” now describes the groups as fixed modern strata, and the Discussion explains this limitation and the role of continuous climate covariates in the fitted models.

Thresholds for pollen counts

The decision to lower the minimum pollen count threshold from 150 to 25 grains per sample raises some questions about data quality. A threshold as low as 25 may introduce significant noise and reduce the reliability of PAPs. The rationale for this decision should be clearly explained and its implications discussed.

Response. We agree that low counts reduce precision and have clarified the procedure. Twenty-five grains is the minimum for an individual sample, whereas 150 grains is the normal rarefaction target. A lower, record-specific rarefaction count is used only where the observed count distribution of that record cannot consistently support 150 grains, avoiding wholesale exclusion of records from data-poor regions. We quantified the coverage consequences of stricter 300-, 500-, and 1,000-grain profiles. These are retention and bias-coverage diagnostics; the complete H1 model inventory was not refitted under the stricter thresholds, and we do not claim threshold invariance.

Changes in the revised article. The revised Methods explain the minimum, target, and record-specific behaviour. Supplementary Result validation presents the pollen-retention analysis, and the Discussion explicitly identifies finite count size, pollen productivity, dispersal, taxonomic resolution, basin characteristics, and rarefaction as limitations.

Figures

Figure 4 shows a lot of information, and the plots are too small. It is difficult to see the major patterns. Furthermore, the panel labels in Figure 2 appear to be misaligned (E and D).

Response. We agree and have reorganised the figures. The study overview remains Fig. 1. The full overview of PAP temporal trajectories is now Fig. 2, the controlled spatial H1 result is Fig. 3, and the spatially controlled temporal H1 result is Fig. 4. The former H2 Fig. 4 has been moved to Supplementary Fig. 1. Quality-control and sensitivity figures are grouped under Supplementary Result validation rather than crowding the main sequence.

Changes in the revised article. Figs. 2–4 now present one major analytical message each at a readable scale, with corrected panel labelling and captions that identify the analysis profile, cohort, controls, and treatment of negative or missing values.

This manuscript provides a novel and valuable analysis of Holocene vegetation dynamics at broad spatial scales. The conclusions are well supported, and my major comments focus on some aspects of the methods and interpretation that require clarification and additional nuance. By addressing these issues, the manuscript will be greatly improved. Again, I recommend publication after these revisions are done.

Response. We thank the reviewer again for the constructive assessment. The requested clarifications, qualifications, figure changes, and validation summaries have been incorporated as described above.

Reviewer #2 (Remarks to the Author)

This paper uses a global dataset of fossil pollen, paleoclimate, archaeologically-attested radiocarbon dates and human indicator pollen types to investigate the relative importance of humans and climate on aspects of pollen diversity and compositional changes from 8.5 cal k BP onwards. Previous work has described changes to various aspects of pollen diversity and compositional turnover measures over multi-millennial timescales across much of the earth surface. Other studies have investigated the relative strengths of relationships between humans versus climate on pollen diversity and compositional turnover for different continents and regions. However, this study represents the first attempt to quantify the relative importance of humans and climate on pollen diversity and compositional turnover measures at a global scale in the Holocene. Using a new method to the field of paleoecology, the authors claim that, at continental and regional

scales, climate was the primary driver of pollen diversity and compositional turnover measures over the period 8 - 2.5cal k BP. This work adds a novel dimension to the conversation (a direct comparison of the strength of drivers globally) and thus represents an interesting and worthwhile contribution to the literature.

Response. We thank the reviewer for recognising the contribution and for identifying areas where the evidence and limitations required clearer presentation. We have revised the response domain to “pollen assemblage dynamics,” fixed the study interval at 8.5–2.0 ka BP throughout, added the PAP trajectories to the main figures, expanded Methods and Discussion, and grouped the new quality-control and sensitivity analyses in Supplementary Result validation.

I do, however, have a number of data presentation and methodological concerns that I would need to see addressed to have confidence in the results presented here. I also see no advantage of including the human event variable here. I understand that the authors would prefer to be able to run these analyses up to the present, but I find the variable reductive and without nuance, it is un-tested and it is much too coarse for it to ever be likely to capture changes to local PAPs. In addition to methodological considerations, I would also like to see an adjustment to the title. In its current form, at least to me, the title suggests that this is a study encompassing the full Holocene, though the primary analyses span the period 8.5 - 2 cal k BP. This is very important because recent work has shown substantial increases in multiple measures of pollen diversity from 2.5 cal k BP onwards across the planet (e.g. in Europe, Giesecke et al, 2019; North America, Stegner and Spanbauer, 2023; global, Mottl et al, 2021 and Gordon et al, 2024), a period of known rapid and major change in human population size, culture, etc. These changes are missed in these analyses given the cut-off date. Climate outweighs human effects on vegetation assemblage dynamics 8 - 2.5 ka BP (or similar) would seem to me to be a more accurate and less ambiguous representation of the analyses.

Response. We agree that the former title overextended the temporal and response-domain scope. The revised title is “Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP,” and the text explicitly states that the results cannot be extrapolated to the last two millennia. We also agree that the event chronology is too coarse to serve as a primary local human-impact predictor. We retain it only as a complementary sensitivity analysis of proxy definition, not as independent confirmation. The SPD is the primary human proxy, and the event analysis is restricted to records and region-age combinations with an observed expert chronology.

Changes in the revised article. The new title, Abstract, hypotheses, Results, Discussion, and Conclusion use the exact interval and pollen-domain wording. Methods and Supplementary Result validation now distinguish SPD-only, event-only, and combined models and explain the limitations of the event chronology.

Data presentation

Though two records are presented in Extended Fig. 3, a key figure missing from this work is a presentation of the full set of PAPs through time for each region. The authors have presented the set of predictor variables, rather than the response variables, which, in my view, are of substantially less importance than the response variables themselves. I find it difficult to make sense of and interpret statistical analyses for which at least some abstraction of the response variables (here, PAPs) are not presented in a paper and thus, for me, these need to be included.

Response. We agree. We now present the temporal patterns of the response variables before interpreting predictor importance. Fig. 2 gives the overview of all PAP temporal trajectories, and the corresponding individual trajectories for all ten PAPs are included in the Supplementary Information. The curves are equal-weighted averages of record-level posterior predictions evaluated only within each record’s observed temporal coverage, so records with longer or more densely sampled sequences do not receive greater weight.

Changes in the revised article. The first Results subsection is now “Temporal patterns in pollen assemblage properties,” with Fig. 2 in the main article and the ten individual PAP trajectories in the Supplementary Information. Methods explains how the trajectories were produced from existing posterior predictions without refitting the temporal models.

Pollen diversity biases

Whether implicitly or explicitly stated, pollen data in studies such as these are used to make inferences on the changing diversity patterns, and relationships with drivers, of the parent vegetative assemblages of pollen samples (as stated by the authors, L652). Thus, the treatment of the pollen data in order to make robust inferences about the actual focus – the parent vegetative assemblage – is of paramount importance. For the reasons outlined below, the treatment of the pollen data in these analyses means that I find it difficult to have full confidence in the robustness of the computed PAPs and thus confidence in the conclusions of this work.

There is substantial variation in pollen productivity among pollen producing plants, which has the capacity to strongly bias estimates of individual PAPs, which the authors acknowledge (L687). Pollen grains from high pollen producing plants are more likely to be deposited and identified in pollen samples than pollen grains from low pollen producing plants. The result is that at low pollen sums, pollen assemblages (samples) are likely to be swamped by pollen grains from a relatively smaller number of high pollen producing plants. This means that the evenness of these samples will be low (high representation by a small number of types), the result of which is a reduced likelihood of detecting pollen types from other pollen producers, which thus artificially depresses the richness of these samples (e.g. Giesecke et al, 2014). Thus at low pollen sums, the richness of samples primarily reflects the evenness of those samples. Similarly, if the likelihood of sampling a diverse set of grains per sample is reduced (given the low evenness of

samples), compositional turnover estimates are likely to be lower for these samples than between samples with higher evenness values. In these analyses, pollen samples are rarefied to either 150 or 25 pollen grains. Even 150 grains is substantially lower than the minimum rarefaction value of ~300-500 grains commonly used in pollen diversity analyses. For those samples rarefied to 25 grains, diversity estimates will almost certainly be affected strongly by these confounding issues. Extended Fig. 3 describes the PAPs and predictor variables for two records, where these described issues can clearly be seen: those Hill numbers N0, N1 and N2 (~richness, ~diversity and ~evenness) are strongly correlated.

These concerns can be at least partly mitigated against by performing some transformation of pollen counts based on relative pollen production and dispersal (etc). There are a number of methods to achieve this, the most notable of which is the REVEALS model, for which there are the requisite datasets for many of the regions used here. However, this is not implemented in these analyses. At the very minimum, I would expect to see a supplementary analysis that considers only the subset of samples and records with a substantially higher minimum pollen grain N (300-500 grains), to check that results are not driven by those issues relating to relative differences in pollen production and dispersal described above. I am of the view that all samples with <150 grains should be excluded from these analyses.

Response. We agree with the reviewer that pollen productivity, dispersal, taxonomic resolution, basin characteristics, and finite count size constrain inference from PAPs. We have therefore changed the title and interpretive wording throughout from vegetation assemblage dynamics to pollen assemblage dynamics and do not treat the PAPs as direct estimates of taxon abundance or parent vegetation diversity. We also clarified that 25 grains is an individual-sample minimum, not a universal rarefaction level. The normal rarefaction target is 150 grains; where a record cannot consistently support that target, a lower record-specific count is determined by its observed sample distribution.

We quantified the coverage trade-off under nominal 300-, 500-, and 1,000-grain thresholds. At 300 grains, 1105 records remain; at 500 grains, 616 remain; and at 1,000 grains, 246 remain. These thresholds change the geographic and temporal cohort substantially. We therefore report them accurately as retention and bias-coverage diagnostics. We did not rerun the complete H1 model inventory at the stricter thresholds and do not claim that the fitted H1 estimates are invariant to pollen-count thresholds.

Correlations among Hill numbers and other PAPs are expected because the metrics share taxa, counts, and mathematical components. We now show the observed correlation structure and compare the complete ten-PAP response with a reduced response on a matched cohort. Among 1,114 complete paired records, the median climate allocation changed from 0.798 to 0.794, retaining the broad climate-over-human ordering while showing that individual contributions can change.

We did not apply REVEALS because globally consistent pollen-productivity estimates, dispersal parameters, basin information, and taxonomic harmonisation are unavailable across the full compilation. Applying it only in selected regions would introduce another major difference in response construction among continents. We now state this inferential limitation directly rather than implying that rarefaction corrects productivity and dispersal bias.

Changes in the revised article. Methods under “Data filtering” and “Pollen richness, diversity, and evenness” now explains the count procedure and expected PAP correlation. The Discussion contains a dedicated pollen-bias paragraph. Supplementary Result validation includes pollen-retention, PAP-correlation, and reduced-response analyses. The article explicitly states that the stricter-threshold work is not a full H1 rerun.

SPD biases

SPDs of archaeologically attested radiocarbon dates are becoming increasingly used to infer relative changes in human paleodemography. However, there is substantial and ongoing debate in the literature among archaeologists about whether this method can actually tell us anything about paleodemography (e.g. Chaput and Gajewski, 2016; Crema and Bevan, 2020; Crema, 2022). It is only material from times and places of archaeological interest that are radiocarbon dated. Thus the number of radiocarbon dated objects through time and space is intensely non-random, clustering around locations and periods of interest to the archaeological community. This is likely the cause of the substantial apparent paleodemographic spikes during transitions between archaeological periods (archaeologists are interested in what caused and was affected by these transitions). On top of this, radiocarbon dating is expensive. So, when possible, other objects – e.g., pottery, coins – are preferentially used to date archaeological remains, which results in a reduction in the number of radiocarbon dates when these objects are readily available. This is most obviously shown in Europe from ~3 cal k BP, from which point the radiocarbon SPD shows a monotonic decline (Extended Data Fig. 4). This apparent reduction is entirely artificial and driven by the general cessation of the use of radiocarbon dating to date archaeological remains, given the proliferation of other dateable material across the continent from this period onwards. Given that this variable underpins this paper’s primary analyses and main conclusions, I would expect some critical discussion in the main text that outlines the well-known and strongly debated limitations of using SPDs for paleodemographic inference, though this is missing. This would then help justify the decision to cut off the analyses at 2.5 cal k BP, which at the moment does not feel sufficiently justified in the text.

Response. We agree that an archaeological SPD is not a direct population curve and have changed the terminology and interpretation accordingly. We describe it as a spatially extensive index of changing dated archaeological activity and discuss how preservation, calibration, research intensity, regional archaeological interest, and changes in dating practice produce non-random coverage. These limitations motivate both the 2.0 ka BP boundary

and our refusal to extrapolate the result to the final two millennia.

Changes in the revised article. The critical discussion is now present in Methods under “Past human presence from archaeological radiocarbon dates” and in the main Discussion. The Introduction and Discussion explain that the common 8.5–2.0 ka BP window was selected to avoid the strongest decline in usable archaeological radiocarbon coverage in parts of Eurasia.

Notwithstanding these comments, of most concern to me is the different spatial scales of the respective variables used in these analyses. The relevant source area of pollen for any given extracted fossil pollen sequence varies from meters up to tens of kilometers (rarely hundreds of km), depending on the deposition environment, location, etc. However, the authors pull radiocarbon dates from either 250km or 500km surrounding a pollen record extraction location to build their SPDs. These scales are vastly different and it is perhaps not surprising that PAPs from, for example, a bog site with a pollen source area in the order of hundreds of meters are not associated with SPDs built from radiocarbon dates pulled from within 250 or 500km but that they are associated with paleoclimate estimates for the actual location of a pollen record. In principle, either (i) a dynamic radius for retrieving radiocarbon dates per pollen record could be used (based on each record’s pollen source area), (ii) these analyses could be run regionally (using averages across PAPs and predictor variables) or (iii), at the very least, a sensitivity analysis should be run that restricts the analyses to those records with the largest pollen source areas, with SPDs built using a radius for radiocarbon dates that is at least equal in order of magnitude to it.

Response. We agree that this scale mismatch cannot be eliminated by a single fixed radius. A dynamic pollen-source radius is not feasible for the global compilation because consistent depositional-environment, basin-size, and source-area estimates are unavailable for all records. The study already includes complementary record-level spatial analyses and regional time-slice analyses. We have now added a matched sensitivity comparison of strict 250 km and 500 km SPD predictors. Among 999 paired spatial records, 963 had two defined zero-truncated balances; the median 500-km-minus-250-km change was 0.025, and 94 records reversed the human-versus-climate ranking. Among 70 paired temporal comparisons, 3 rankings reversed. These results show that the radius can matter for individual estimates and should not be treated as a precise match to a pollen-source area.

Changes in the revised article. Methods now explains the proximity–coverage trade-off between 250 km and the 500 km fallback. Supplementary Result validation reports the matched radius analysis, and the Discussion treats the scale mismatch as a limitation of individual-record inference.

Human events variable

Even by itself, the development of this variable represents a major undertaking and whilst I appreciate the substantial effort, I find the decomposition of continuous human change into categorical events represented by a binary variable overly simplistic and almost certain not to capture diverse changes across the PAPs considered here. I also find there to be a lack of description and justification of the method. Which experts were asked? Why were these experts asked? Who do they represent? Why and how were these events chosen? By whom? What were the exact criteria used for each event * region considered? Indeed, for Central and South America, the authors use an established index from the literature, which underlines the extent of the work required for the development of indices such as these for individual continents, let alone for all continents as is attempted here.

The reductive nature of the index is exemplified best for North America, for which the authors provide three event categories: No Evidence of Impact, First Cultivation and European Settlement. For many Indigenous North American peoples, cultivating domesticated plant types remained supplementary to a much broader focus on foraging wild foods (both plant and animal) over the Holocene. Wild foods were often foraged with the substantial assistance of fire, which was used extensively across the continent before, during and after cultivation of domestic crops. According to this index, there was no human impact in North America before First cultivation, which is certainly untrue. Indigenous groups across North America have impacted vegetative communities over the entire Holocene and into the Pleistocene both directly, through burning, and indirectly, through the removal of the largest animals. Both of these would have strongly impacted vegetative communities, but their effects are entirely missed by the coarse categories here. It would be a vast amount of work to develop a pollen-based index of human impact for all regions globally that was transparent and not overly reductive. I do not believe that, in its present form, the inclusion of this binary variable adds to this work and I would simply remove it and curtail the analyses at the end of the period of SPD relevance, however defined.

Response. We agree that the binary event chronology is coarse and cannot represent the continuous, diverse, and locally specific activities described by the reviewer. We have removed “No Evidence of Impact” as an interpretation of human absence: a non-detected event now means only that the specified pollen criterion was not detected. SPD is the primary human proxy. The event chronology is retained solely to test whether the broad result depends on replacing or supplementing SPD with an expert-screened, pollen-derived proxy. Because this proxy partly shares its evidence source with the PAP response, we describe the comparison as sensitivity to proxy definition, not independent validation.

In the matched observed-event cohort, the spatial comparison contained 151 records and the temporal comparison contained 61 region-age combinations. The combined and event-only estimates changed in both directions relative to SPD-only models, confirming that attribution depends partly on proxy definition. We have expanded the regional criteria and

construction procedure in Methods and the Supplementary tables. The final submission will also provide complete expert names, regional roles, selection rationale, and source attribution once confirmed by all authors; we will not invent this provenance where it is not yet documented in the repository.

Changes in the revised article. The event variable has been demoted from co-equal evidence to a complementary sensitivity analysis. Methods now defines the event construction and regional criteria, the Discussion explains its coarseness and partial circularity, and Supplementary Result validation compares SPD-only, event-only, and combined models on matched observed-event cohorts.

Statistical analyses

Firstly, I cannot see a summary of the total variance explained by the models here. This is of crucial importance because if the models explain a very small amount of the total variation in the data, then it would be difficult to say that humans/climate are most/not important drivers of the PAPs, in my view. These should be added as a Supplementary table and summarised in some way - perhaps across records, by region - as an Extended Data Figure.

Response. We agree. The revised article separates total adjusted R^2 from the relative allocation of explained variation. Of the 1,262 records in the original spatial SPD H1 inventory, 1,156 had an estimable HVarPart decomposition, with mean total adjusted R^2 of 0.352. The controlled spatial reporting cohort had median total adjusted R^2 of 0.450. Distributions are shown rather than relying only on these summaries because fit varies substantially among records.

During this work we also identified and corrected the extraction of individual HVarPart contributions. We now retain signed hierarchical contributions, show a zero-truncated composition only as a display, and separately report a profile excluding models with negative individual components. The broad climate-over-human ordering is retained across these profiles, but the numerical allocations are not treated as interchangeable with each other or with total model fit.

Changes in the revised article. Total adjusted R^2 is reported in the main H1 Results and in Supplementary tables and figures showing its distribution and relationship with predictor importance. Methods and Supplementary Result validation define the corrected HVarPart profiles, unique adjusted- R^2 fractions, and shared commonality components.

Secondly, temporal and spatial autocorrelation do not seem to be accounted for in these models, and their effects not checked for. These effects potentially confound model inferences and thus either an explanation as to why they were not included needs to be added to the methods, or some additional variables need to be added to the models. This is likely of most importance to

those analyses in Fig 2 where there will undoubtedly be a substantial trend component to the PAPs over the long period considered.

Response. We agree that dependence must be controlled and diagnosed. The primary spatial H1 figure now comes from models that include time as a control group within each record. The primary temporal H1 figure uses selected distance-based Moran eigenvector maps to represent spatial structure among records within regional time slices. We additionally report residual Moran tests before and after spatial control, temporal residual diagnostics, spatial thinning at 250 and 500 km, leave-one-region and leave-one-climate-zone checks, and unique-fraction and reciprocal-control summaries. These diagnostics retain the broad aggregate ordering while showing residual dependence and instability in individual estimates.

Changes in the revised article. The controlled analyses are now the main H1 Figs. 3 and 4. Methods explains time and dbMEM controls, while the complete dependence, thinning, and leave-out procedures and results are grouped under Supplementary Result validation.

References

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Stegner, M.A. and Spanbauer, T.L. (2023) 'North American pollen records provide evidence for macroscale ecological changes in the Anthropocene', *Proceedings of the National Academy of Sciences*, 120(43), p. E2306815120.

Response. We thank the reviewer for these references. We have used them to strengthen the temporal framing and the discussion of SPD and pollen-diversity limitations. The final cited subset will be checked with the co-authors so that each retained reference directly supports the revised text and the regional coverage remains balanced.

Reviewer #3 (Remarks to the Author)

I work on Quaternary paleoclimate reconstructions and with Earth system scientists, and can comment on the relevance and impact of this study. It is indeed assumed by many that humans have played a significant and increasing role in shaping Holocene vegetation assemblage dynamics for the 8.5 to 2 ka BP study interval. This assumption is relevant, for instance, to the vigorous debate about whether the Anthropocene should be viewed as having gradually unfolded over tens of millennia (the “Anthropocene event” of Gibbard et al., 2022; see also the gradualist depiction in fig. 1 of Ellis, 2023), or whether significant impacts are restricted to more recent times. The conclusion that humans were not responsible for broad-scale ecological processes during this interval of study is therefore important to this debate, and it sets a baseline for human-driven impacts that were to later overwhelm Earth system functioning.

The study itself is based on the novel statistical analysis of two very large and independent data sets. The conclusions seem to be well supported by the data, and careful thought is given to explaining their limitations. There is one controversial aspect of Holocene climate that is not addressed but perhaps could be. It concerns claims by Ruddiman et al. (e.g. 2003, 2007, 2020) that the emergence of intensive rice cultivation in eastern Asia was responsible for the release of significant carbon dioxide and methane into the atmosphere, from 7 ka and 5 ka respectively. This would explain why global temperatures remain relatively flat even as summer insolation at 65°N declined from its peak in the Early Holocene. These claims have been questioned by Zalasiewicz et al. (2024) who noted that isotopic and modelling studies are consistent with oceanic outgassing rather than anthropogenic causes for these rises in greenhouse gases. Moreover, Summerhayes et al. (2024) noted that global average insolation may have in fact risen slightly over the past 7 kyr, potentially leading to the degassing of the global ocean which is by far the largest CO₂ reservoir on the planet. The question then, as to whether there is evidence within the pollen record for substantial cultivation of rice in eastern Asia, becomes relevant in assessing the extent of human impacts on the Earth system.

Response. We thank the reviewer for identifying this broader context. We agree that the timing and magnitude of eastern-Asian rice cultivation is important to the early anthropogenic hypothesis. The present analyses, however, cannot test that hypothesis: the response variables are ten broad PAPs rather than harmonised rice-pollen occurrence or land-use

reconstructions, and the predictor design does not estimate greenhouse-gas forcing. Adding such an analysis would require taxon-level harmonisation, eastern-Asian archaeological and cultivation predictors, and validation designed for that causal question. We have therefore not inferred the presence, extent, or climatic effect of rice cultivation from the PAP results, but have added the question explicitly as an inferential boundary and direction for future work.

Changes in the revised article. The final Discussion paragraph now explains that a taxon-specific reconstruction of eastern-Asian rice cultivation and early anthropogenic greenhouse-gas forcing would require a different response domain and inferential design and is not a sensitivity analysis of the present models.

I consider the MS to be publishable after minor revision, and perhaps following consideration of the question of eastern Asia rice cultivation. The Holocene is crucial to our understanding of the rise of civilizations and the emergence of the modern world. This MS will have a profound impact on the way we approach these fundamental issues.

Martin J. Head

5 July 2025

Response. We thank the reviewer for this encouraging assessment. We have considered the eastern-Asian rice-cultivation question and clarified why the present PAP-based analysis cannot resolve it, while ensuring that the revised article does not make a taxon-specific or greenhouse-gas attribution.

References

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Response. We thank the reviewer for providing this literature. The revised Discussion recognises the question but does not use these papers to imply that our PAP models test competing mechanisms of Holocene greenhouse-gas change.

Minor detailed comments on the MS

Line 69. Should there be consistency here in the precision, i.e. 8.5–2.0 ka BP? See Line 336 where this consistency is maintained.

Response. Agreed. We now use 8.5–2.0 ka BP consistently when specifying the study interval.

Line 82. For land use, reference to Our World in Data would be better: <https://ourworldindata.org/land-use>.

Response. Thank you for the suggestion. We revised the surrounding text and will make the final reference choice together with the co-authors, retaining this source only where it directly supports the statement made.

Line 95. not “middle Holocene” but “Middle Holocene” if you are referring to the formal substage/subepoch (8.2–4.2 ka), or “mid-Holocene” if more loosely.

Response. Corrected. We use “Middle Holocene” for the formal subdivision and “mid-Holocene” for informal usage.

Line 113. Pleistocene and Holocene epochs (not periods).

Response. Corrected to “epochs.”

Line 116. “the term “period” not needed here because the Holocene can only refer to a unit of time.

Response. Corrected. We removed the unnecessary use of “period” with Holocene.

Line 124. reflects, not reflect (“each” refers to the singular).

Response. Corrected to “reflects.”

Line 134. “on the period between 8.5 and 2.0 ka BP” if the consistency in precision is justified.

Response. Corrected, with the study interval consistently expressed as 8.5–2.0 ka BP or “between 8.5 and 2.0 ka BP,” according to sentence structure.

Line 147. we would expect an altered covariance between the PAPs within different climate zones on continental regions.

Response. Corrected and clarified. H2 is now framed as a prediction about changes in relationships among PAPs within continental and subcontinental groups, rather than as a taxon-specific vegetation response.

Line 150. This subheading is a little confusing because while it references H1, it describes the result rather than the stated hypothesis. I realize that restating the hypothesis could be misleading here, so perhaps frame the hypothesis as “H1: Holocene palaeoclimate versus human impact”.

Response. Agreed. We revised the subheading to “H1: Holocene human impact secondary to palaeoclimate,” which identifies the hypothesis and the observed result while retaining the temporal context.

Line 157. Middle to Late Holocene – you are referring to official subseries of the Holocene, so the first initial of the modifiers “middle” and “late” must be capitalized. If this is not done, you will need to define what will then be informal terms.

Response. Corrected to “Middle to Late Holocene” where the formal subdivisions are intended.

Line 175. Again an issue of consistency in precision. In Asia 3.5 ka BP so for Europe should that be 3.0 ka BP?

Response. Corrected. Ages are now reported with consistent precision within each comparison.

Line 218. palaeoclimate data between 8.5 and 2.0 ka BP to investigate ... (then delete “8.5 and 2.0 ka BP” at the end of the sentence).

Response. Corrected and the duplicated interval was removed.

Line 220. Our first hypothesis posited that ...

Response. Corrected. The sentence now begins with an explicit formulation of the first hypothesis.

Line 241. PAPs not PAPS.

Response. Corrected throughout to “PAPs.”

Line 275. “mid” is not a stand-alone word in most dictionaries, so it would be “mid-Holocene” as an informal interval of time.

Response. Corrected to “mid-Holocene.”

Line 326. “factors like” better as “such factors as”.

Response. Corrected to “such factors as.”

Line 336. “from 8.5 to 2.0 ka BP”. Use an en-dash when referring to, but not specifying, a range, such as “over the range 8.5–2.0 ka BP”. I note here that “2.0 ka unlike elsewhere is states to two significant figures, which seems appropriate precision if supported by the dates themselves.

Response. Corrected. We use “from 8.5 to 2.0 ka BP” in prose and an en dash in the adjectival or compact range “8.5–2.0 ka BP.”

Line 467. You do make it clear that Eric C Grimm is no longer with us, but are you perhaps dedicating this MS to the memory of the person rather than the person themselves? Remember to change “manuscript” to “article” when revising the MS.

Response. Corrected. The dedication now refers to the memory of Eric C. Grimm, and submission-facing prose uses “article” rather than “manuscript” where appropriate.